

HARDWARE TROUBLESHOOTING & DIAGNOSTICS

1. Common Desktop PC Hardware Issues

- **RAM Failure or Improper Seating:**

Symptom: PC powers on but displays a blank screen, accompanied by continuous long beeps or repeating short beeps from the motherboard speaker.

- **Failing Storage Drive (HDD/SSD):**

Symptom: Frequent Blue Screen of Death (BSOD) crashes, audible clicking noises (for HDDs), or a "Boot device not found" error during startup.

- **Overheating CPU (Thermal Throttling/Trip):**

Symptom: The system boots normally but abruptly shuts down after a few minutes under heavy load, often preceded by excessively loud CPU fan noise.

- **Dead CMOS Battery:**

Symptom: The BIOS loses its time and date settings upon losing AC power, and prompts a "CMOS Checksum Error" on boot.

- **Failing Power Supply Unit (PSU):**

Symptom: Random, unexplained restarts or shutdowns, or the PC is completely unresponsive when the power button is pressed (no lights, no fan spin).

2. Laptop Not Charging Troubleshooting Plan (Basic Tools)

Scenario: Laptop is not charging even when plugged in.

1. **Check the Power Outlet:** Plug another device (e.g., a lamp or phone charger) into the same wall socket to confirm it is actively providing power.
2. **Inspect the Power Adapter & Cable:** Visually check the entire length of the cable for physical damage, frayed wires, or sharp kinks. Check the connector tip for bent pins or debris, and smell the brick for any burnt plastic odors.
3. **Examine the Laptop's DC Jack:** Look into the charging port on the laptop with a flashlight. Check for lint, debris, or damage. Gently wiggle the inserted plug to see if the internal jack is loose or broken from the motherboard.
4. **Perform a Hard Reset / Power Drain:** Disconnect the AC adapter and remove the laptop battery (if externally removable). Press and hold the power button for 30–60 seconds to discharge residual static electricity. Reconnect only the AC adapter and attempt to power on.

5. **Test with Alternate Hardware:** If available, test the laptop with a known-working, compatible charger. Alternatively, test the suspect charger on another identical/compatible laptop to isolate whether the fault lies with the adapter or the laptop itself.

3. USB Phone Charger Multimeter Test

Procedure: The multimeter is set to DC Voltage (20V range). The black probe is connected to the Ground (GND) pin and the red probe is connected to the Voltage (VCC) pin of a USB breakout board plugged into the charger.

Simulated Measurement: $V_{out} = 5.08V$

Conclusion: The measured voltage is 5.08V. This is well within the standard USB specification expected range of 4.75V to 5.25V, indicating that the charger's voltage output is stable and functioning correctly.

4. Power Supply 12V Rail Multimeter Test

Scenario: PC is randomly shutting down. Testing the 12V rail on a disconnected ATX connector.

Procedure:

1. Disconnect the 24-pin ATX connector from the motherboard (ensure the PSU is unplugged from the wall during this step).
2. "Jump" the PSU to turn it on by bridging the PS_ON pin (usually a Green wire, pin 16) to any Ground pin (Black wire) using a paperclip, then plug the PSU back into the wall.
3. Set the multimeter dial to DC Voltage (typically marked as V_{DC} or VDC) in the 20V range.
4. Insert the black multimeter probe into any Ground pin (black wire) on the ATX connector.
5. Insert the red multimeter probe into the 12V rail pin (yellow wire) to read the voltage.

Simulated Measurement: $V_{12V} = 11.23V$

Diagnosis: The voltage reads 11.23V. The ATX specification dictates that the 12V rail must remain within a $\pm 5\%$ tolerance, meaning the acceptable range is 11.40V to 12.60V. Since the voltage drops below 11.5V, this indicates a **failing Power Supply Unit (PSU)**. The components (especially the CPU and GPU) are being starved of adequate stable voltage under load, triggering the hardware protection circuits and causing the random system shutdowns.